XRPL Validator — Daily Engineering Report · 2026-09-27

Period: 2026-09-27 00:00 to 24:00 US Eastern (2026-09-27 04:00 to 2026-09-28 04:00 UTC); ledgers #107,262,588 to #107,284,759. Every count from the ledger and the validator's logs stops at #107,284,759. Generated 2026-10-21 00:47 EDT by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Revised on 2026-10-21 in report format v3; the original is kept in this repository's history. Built only from records that exist for this day. Available: the day's first and last ledger, from a public full-history server; signing records; Step 1 receipt records (they begin 2026-08-29); deployment records (they begin 2026-09-11); xrpl-validator merge records (they begin 2026-07-04); amendment data (carried from the original report). Not available: ledger headers (the reference node holds ledgers from 2026-10-04 22:52 UTC); Steps 2–5 receipt records (they begin 2026-10-09); data.xrpl.org agreement scores (scored days begin 2026-04-04; some days have none).

Status

Normal (agreement not checked) — coverage outside planned maintenance 100.000% (objective 99.9%), agreement with the network not checked (the reference node no longer holds these ledgers' hashes), no open engine receipt.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** signed 22,163 of the 22,172 ledgers the network closed (99.959%) (agreement not checked: the ledgers' hashes are no longer held). Unsigned: 9 during planned restarts.
- **Native engine:** 3 engine receipts recorded (3 fixed, 0 explained, 0 without a recorded fix).
- **Changes:** 1 deployment (cycle 158), 5 commits merged into main, 3 engine findings fixed (F403–F405).

Service levels

Measure	Value
Ledgers closed by the network	22,172
Validations issued	22,163
Coverage (validations / ledgers closed)	99.959%
Validations matching the network's validated ledger hash	not checked: the reference node no longer holds these ledgers' hashes
Unsigned during planned restarts	9
Unsigned outside maintenance	0
Coverage outside planned maintenance (objective 99.9%)	100.000%

Measure	Value
Restarts	1
Basis: the hash the first peer announced as accepted (before cycle 183)	22,163

Restarts:

- 12:00:57 UTC: deployment of cycle 158; validations resumed at #107,269,990 (9 ledgers unsigned between the last validation before the restart and the first after it)

The basis rows sum to validations issued. Coverage counts the validations the validator issued for ledgers the network closed in the period. Agreement compares each validation's ledger hash with the network's validated hash for that ledger. data.xrpl.org scores UTC days from the validations its crawlers received, so it can differ.

Native engine conformance

Ledgers with a receipt in the period (#107,262,588–#107,284,759), by step:

Step	Ledgers with a receipt
Step 1 · native engine	3
Step 2 · records	0
Step 3 · order	0
Step 4 · peer data	0
Step 4 · validations	0
Step 5 · own hash	0
Writer	0

Live counters are kept only for the current run; none exist for a past day.

Receipt register

ID	Time	Ledger(s)	Step	Description	Disposition
R-01	—	#107,263,299	Step 1 · native engine	1 object(s) differed in content	fixed: F403, commit e7d7655, deployed in cycle 158 (2026-09-27 12:00 UTC)
R-02	—	#107,266,300	Step 1 · native engine	1 transaction result(s) differed from rippled's; 1 ledger change(s) the network made were absent from ours: 3852349A; 1 object(s) differed in content	fixed: F404, commit ae84a26, deployed in cycle 158 (2026-09-27 12:00 UTC)
R-03	—	#107,278,773	Step 1 · native engine	1 transaction result(s) differed from rippled's; 6 change(s) of ours the network did not make; 1 object(s) differed in content	fixed: F405, commit 51b0cbe, deployed in cycle 159 (2026-09-28 14:08 UTC)

A receipt is recorded whenever the validator's own work differs from the network's, or the network's trusted validations reached it short of quorum. One ledger flagged by several steps is one receipt. Dispositions: fixed (with the finding, commit and deployment), explained (not a defect, with the reason), or open. Receipt and incident IDs are numbered per report: R-01 here is R-2026-09-27-01, INC-01 is INC-2026-09-27-01. Times are not shown where the reference node no longer holds the ledger.

Incidents

No incidents.

Change log

Deployments

Cycle	Main	Deployed	Test suites	Replay windows	Post-deploy health	Soak
158	`ae84a26`	12:00:57 UTC	n/a	n/a	410 matched, 0 mismatched	#33 from 12:28 UTC: ended early: superseded by the next deployment (2026-09-28 14:35 UTC)

Cycle 158 delivered: `02001fb` feat(xrpl-node): /api/connections and /api/amendments for the ops copilot; `40d8d4a` feat(xrpl-node): vote for the pending amendments the engine is ready for; `e7d7655` fix(xrpl-ledger): F403 — an IOU Clawback moves the exact amount through rippleCredit; `ae84a26` fix(xrpl-ledger): F404 — a ticket-funded XRP Payment's reserve excludes the spent ticket

Each deployment passes the unit and port test suites and replays mainnet history windows with zero mismatches before it ships, is health-checked after it starts, and then soaks for 24 hours.

Merged into main

Merged	Repo	Commit	Subject
11:10 UTC	xrpl-validator	`e7d7655`	fix(xrpl-ledger): F403 — an IOU Clawback moves the exact amount through rippleCredit
11:10 UTC	xrpl-validator	`ae84a26`	fix(xrpl-ledger): F404 — a ticket-funded XRP Payment's reserve excludes the spent ticket
11:10 UTC	xrpl-validator	`40d8d4a`	feat(xrpl-node): vote for the pending amendments the engine is ready for
11:10 UTC	xrpl-validator	`02001fb`	feat(xrpl-node): /api/connections and /api/amendments for the ops copilot
21:47 UTC	xrpl-validator	`51b0cbe`	fix(xrpl-ledger): F405 — an AMM withdrawal by the issuer checks the reserve for its own currency

Findings fixed

- **fix(xrpl-ledger): F403 — an IOU Clawback moves the exact amount through rippleCredit** (`e7d7655`, xrpl-validator) Mainnet #107263299 DC750F7C1704: an issuer clawed 200,000 XXPAYMENTS back from a line holding 200,000.0000000005, leaving 0.0000000005. rippled's Clawback (applyHelper) moves min(accountHolds, Amount) from the holder to the issuer with directSendNoFee: exact decimal amounts and rippleCredit's line mechanics. The engine parsed both amounts as f64, subtracted, and wrote 0.00000000049476... — f64 cannot hold …
- **fix(xrpl-ledger): F404 — a ticket-funded XRP Payment's reserve excludes the spent ticket** (`ae84a26`, xrpl-validator) Mainnet #107266300 F31123B49AFE: a 5,877-drop XRP Payment paid for with a Ticket, from an account holding 3.002177 XRP at OwnerCount 10 (ten tickets). rippled's Transactor::apply deletes the ticket

being spent (consumeSeqProxy: OwnerCount 10 -> 9) before Payment::doApply's funding guard, mPriorBalance < amount + max(reserve, fee): the reserve is 1 + 0.2 x 9 = 2.8 XRP and mainnet succeeded. The engine makes that …

- **fix(xrpl-ledger): F405 — an AMM withdrawal by the issuer checks the reserve for its own currency** (`51b0cbe`, xrpl-validator) Mainnet #107278773 784FE06097CB: a tfWithdrawAll from an XRP/001 pool by rJENNivB, the account that issues 001, holding 1.41 XRP at OwnerCount 2. rippled's AMMWithdraw sufficientReserve (fixAMMv1_2) asks only whether keylet::trustLine(account, issue) exists, and an issuer never has a line to itself, so the check runs for the issuer's own currency too: a third object needs 1 + 0.2 x 3 = 1.6 XRP, and with the pool's …

Other engineering changes

- **feat(xrpl-node): vote for the pending amendments the engine is ready for** (`40d8d4a`, xrpl-validator) The vote list followed rippled 3.2.0's supported set, not the engine. It voted for XChainBridge and fixXChainRewardRounding — the XChain transactors are a blind source port with no specimen, and the rounding fix is not gated in them — and not for PermissionDelegationV1_1, which the engine implements and has verified (finding 360; devnet campaign 22, all byte-exact). James, 2026-09-26: vote accordingly. 45 votes from …
- **feat(xrpl-node): /api/connections and /api/amendments for the ops copilot** (`02001fb`, xrpl-validator) /api/connections reports which endpoints the validator's ledger feed and its RPC client are on ("primary", or the public fallback's URL — the primary's own URL is never echoed), how long the current fallback stint has lasted, stints and failovers since start, and each validation relay slot: the hub holding its session and for how long, sessions opened, failures in a row and the last failure. ws_sync::client() is the process's one …

Independence program

No independence record for UTC day 2026-09-27; records begin 2026-10-20.

Network

No ledger records for this day: the reference node holds ledgers from 2026-10-04 22:52 UTC (#107,436,647).

Amendment watch

Carried over unchanged from the original report, which read it from the XRP Ledger when it was generated; the reference node no longer holds this day's ledgers.

Read from the XRP Ledger: the Amendments object as of the last ledger of the day, and the day's EnableAmendment pseudo-transactions.

Amendment	Majority since	Activation due
PermissionDelegationV1_1	2026-09-24 21:25 UTC	2026-10-08 21:25 UTC
fixBatchV1_2	2026-09-25 14:12 UTC	2026-10-09 14:12 UTC
BatchV1_1	2026-09-25 14:46 UTC	2026-10-09 14:46 UTC

Amendments enabled on mainnet: 94.

Open items

None.

Notes

Hand-written notes on the day's XRPL work: research and operations.

- **Soak #32: two engine receipts, both fixed.** Soak #32 of build c79ee89 (from 13:21 EDT on 2026-09-26) recorded two engine receipts, F403 and F404 below; it ended with this morning's deployment at 08:01 EDT. Over the soak the shadow compared 16,823 ledgers and the state-hash check matched 16,828, with no state-hash mismatch: the receipts are differences in the native engine's own results, found by its comparison against libxrpl, not in the validator's state.
- **Finding F403 (Clawback of an issued token).** At ledger 107,263,299 an issuer clawed back 200,000 units of its token from a trust line holding 200,000.0000000005. rippled moves the smaller of the holder's balance and the amount with exact decimal arithmetic and left 0.0000000005 on the line; the engine did the subtraction in binary floating point, which cannot represent 200,000.0000000005, and wrote about 0.00000000049476. Clawback now moves the amount through the engine's exact trust-line credit, the one payments use, which also brings rippled's line bookkeeping. The mainnet transaction is pinned as a regression vector. Fixed in e7d7655.
- **Finding F404 (XRP payment paid for with a Ticket).** At ledger 107,266,300 an account holding 3.002177 XRP with ten tickets (owner count 10) sent 5,877 drops using one of them. rippled deletes the ticket being spent before it checks the payment against the reserve, so the reserve was $1 + 0.2 \times 9 = 2.8$ XRP and mainnet accepted the payment; the engine checked earlier, still counting the ticket (3.0 XRP), and refused it. The same pattern as F402 (TicketCreate, 2026-09-25). Fixed in ae84a26. Two other transaction types, EscrowCreate and TrustSet, also check their reserve after the ticket is consumed in rippled while the engine checks before; no mainnet transaction has exercised the difference, and devnet transactions will be generated to test both.
- **Deployment.** Build ae84a26 replaced the live build at 08:01 EDT, after its gate (1,691 suite tests and 745 flow-engine tests passing) and both mainnet replay windows (60 and 98 ledgers, zero mismatches), by warm restart: it rebuilt its state in 22.4 seconds and resumed two ledgers behind the network. The health check matched 410 ledgers with no mismatches, and soak #33 began at 08:28 EDT (seed ledger 107,269,979). Besides F403 and F404 it adds two read-only status endpoints for the operations monitor (the validator's connections, and its amendment votes) and changes the votes.
- **Amendment votes.** The validator now votes for exactly the pending amendments its engine implements and has verified: PermissionDelegationV1_1 is added (implemented and verified against devnet), XChainBridge and fixXChainRewardRounding are removed (the engine's cross-chain bridge code has never run against a real bridge). 45 votes instead of 46. The first voting ledger after the deployment, 107,270,143, carried the 45 votes and reached the reference node.
- **Amendment readiness.** The operations monitor now warns, with the date, when an amendment with majority on mainnet is one the engine is not ready for. Today that is fixBatchV1_2, due about 2026-10-09 14:12 UTC; its source is still unpublished.

Corrections to earlier reports

None.

XRPL research

Automated research — candidate amendment ideas

Ideas generated and prior-art-checked by the XRPL Amendment Board, an automated research tool (read-only board: iceland.halcyon-names.io). They are exploration leads, not proposals.

- **On-Ledger Asynchronous Multisig Proposals** (custody) Add a `PendingTxn` ledger object (`ltPENDING_TXN`, owner-reserved by the multisig account, `keylet::pendingTxn(account, sfSequence)`) that holds one s
- **Minimum-Fill Offers to Stop Dust Execution** (dex) Add an optional `sfMinFillAmount` (STAmount, in the offer's TakerGets currency) to `OfferCreate` and the `Offer` SLE. In `BookStep`, a crossing that w
- **Tranche Vesting Escrows in One Object** (payments) Extend `EscrowCreate` and the `Escrow` SLE with three optional fields: `sfVestingInterval` (UInt32 seconds), `sfVestingPeriods` (UInt8, 2–48), and `sf

Appendix — commits

Committed	Merged into main	Repo	Commit	Subject
09-27 05:29	09-27 11:10	xrpl-validator	`e7d7655`	fix(xrpl-ledger): F403 — an IOU Clawback moves the exact amount through rippleCredit
09-27 08:12	09-27 11:10	xrpl-validator	`ae84a26`	fix(xrpl-ledger): F404 — a ticket-funded XRP Payment's reserve excludes the spent ticket
09-26 20:18	09-27 11:10	xrpl-validator	`40d8d4a`	feat(xrpl-node): vote for the pending amendments the engine is ready for
09-26 19:43	09-27 11:10	xrpl-validator	`02001fb`	feat(xrpl-node): /api/connections and /api/amendments for the ops copilot
09-27 21:42	09-27 21:47	xrpl-validator	`51b0cbe`	fix(xrpl-ledger): F405 — an AMM withdrawal by the issuer checks the reserve for its own currency

Times in UTC. A commit appears on the day it was committed to main or merged into it.